Efficiently explaining Constraint Satisfaction Problems (CSP) with unsat core optimisation

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Motivation

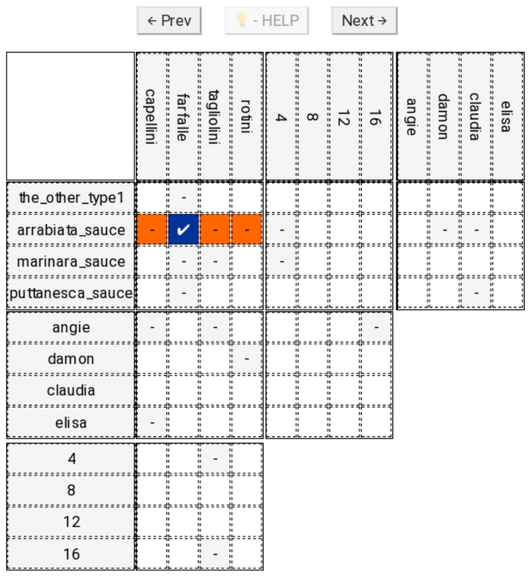
"From *human-level* problem specification, to *human-level* solving and explanations."

ZEBRA TUTOR: EXPLAINING LOGIC GRID PUZZLES

ZebraTutor is an end-to-end solution for solving logic grid puzzles and for explaining, in a human-understandable way, how this solution can be obtained from the clues. Here is an example puzzle. The computer has already solved the problem ! But can you solve it ?

*Click **prev** and **next** to navigate in the human-like solving process.*

PUZZLE



CLUES

1. The person who ordered capellini paid less than the person who chose arrabiata sauce
2. The person who ordered tagliolini paid more than Angie
3. The person who ordered tagliolini paid less than the person who chose marinara sauce
4. Claudia did not choose puttanesca sauce
5. The person who ordered rotini is either the person who paid $8 more than Damon or the person who paid $8 less than Damon
6. The person who ordered capellini is either Damon or Claudia
7. The person who chose arrabiata sauce is either Angie or Elisa
8. The person who chose arrabiata sauce ordered farfalle
9. Logigram Constraint
   - Transitivity constraint
   - Bijectivity
   - Combination of logigram constraints

LEGEND

New fact · Known fact · Nested Explanation · Countradiction Unsatisfiable · True · False · Contradiction Fact

https://bartbog.github.io/zebra/

Step-Wise Explanations for CSPs

$\mathbf{I}_1$ = $\mathbf{I}_0$ (initial information) + $\mathbf{N}_0$ (new information in the grid)

C *constraints (alldifferent, George did not take pasta, …)*

How did we get there ?

1. Using one or a subset of the constraints ($\mathbf{S} \subseteq C$)
2. Part of the already derived Facts ($\mathbf{E} \subseteq I$)

EXPLANATION

Explanation is an implication of the form: $\mathbf{E}_i \wedge \mathbf{S}_i \Rightarrow \mathbf{N}_i$

EXPLANTION SEQUENCE

$\mathbf{I}_0 \xrightarrow[\mathbf{S}_0]{+\mathbf{N}_0} \mathbf{I}_1 \xrightarrow[\mathbf{S}_1]{+\mathbf{N}_1} \mathbf{I}_2 \xrightarrow[\mathbf{S}_2]{+\mathbf{N}_2} \dots \xrightarrow[\mathbf{S}_n]{+\mathbf{N}_n} \mathbf{I}_{end}$

[ECAI2020]

Efficient generation

Figure inspired by [Saikko 2016]

1 **What are good or bad explanations ?**

Use cost function f for quantifying difficulty

How do we guarantee optimality?

2 **How do I get to $\mathbf{I}_{end}$ efficiently?**

Focus search on finding the next cost-optimal explanation at any point in the explanation sequence.

3 **CONSTRAINEDNESS** ***We want to explain 1 fact at a time !***

INCREMENTALITY **What information can we re-use ?**

Every subsequent call to finding an explanation is similar to the previous one

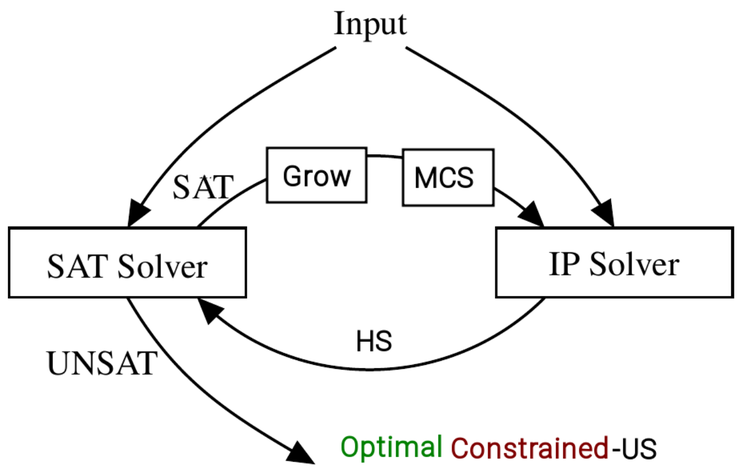


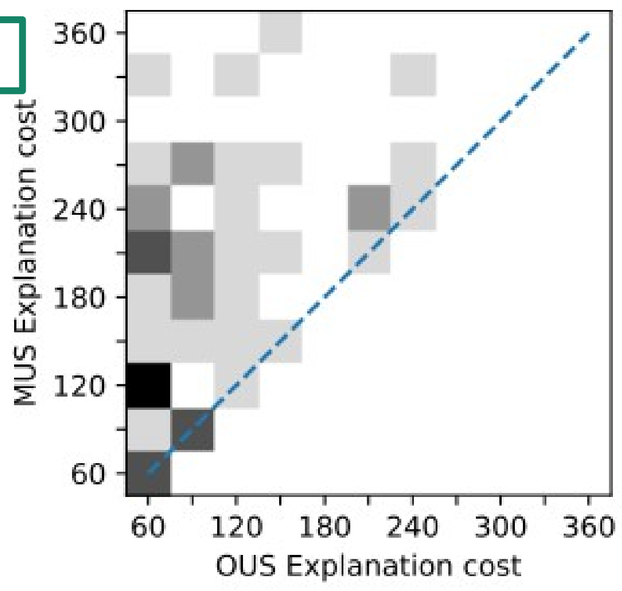
Figure: Optimal constrained unsatisfiable subset computation

MUS *[ECAI2020]* → ***OUS*** (*+ optimality*) → ***OUS+I*** (*+incrementality*) → ***OCUS*** (*+constrainedness*) → ***OCUS+I*** (*+incrementality*)

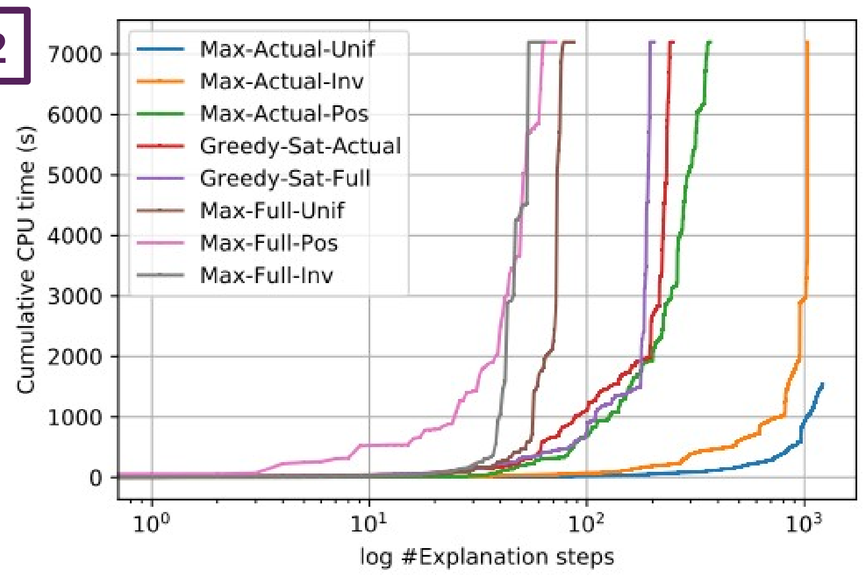
Experimental Results

[Accepted IJCAI 21]

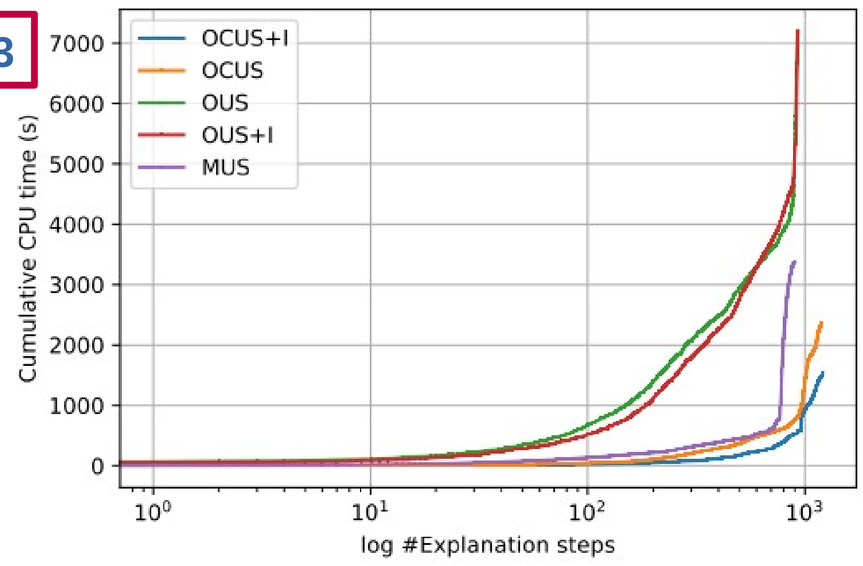
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[Accepted IJCAI 21] Emilio Gamba, Bart Bogaerts and Tias Guns "*Efficiently Explaining CSPs with Unsatisfiable Subset Optimisation*"
[ECAI2020] Bogaerts, Bart, et al. "Step-wise explanations of constraint satisfaction problems." 24th European Conference on Artificial Intelligence (ECAI). 2020.
[AIJournal] Bart Bogaerts, Emilio Gamba and Tias Guns "A framework for step-wise explaining how to solve constraint satisfaction problems"
[Saikko 2016] Saikko, Paul, Johannes Peter Wallner, and Matti Järvisalo. "Implicit Hitting Set Algorithms for Reasoning Beyond NP." KR. 2016.

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